

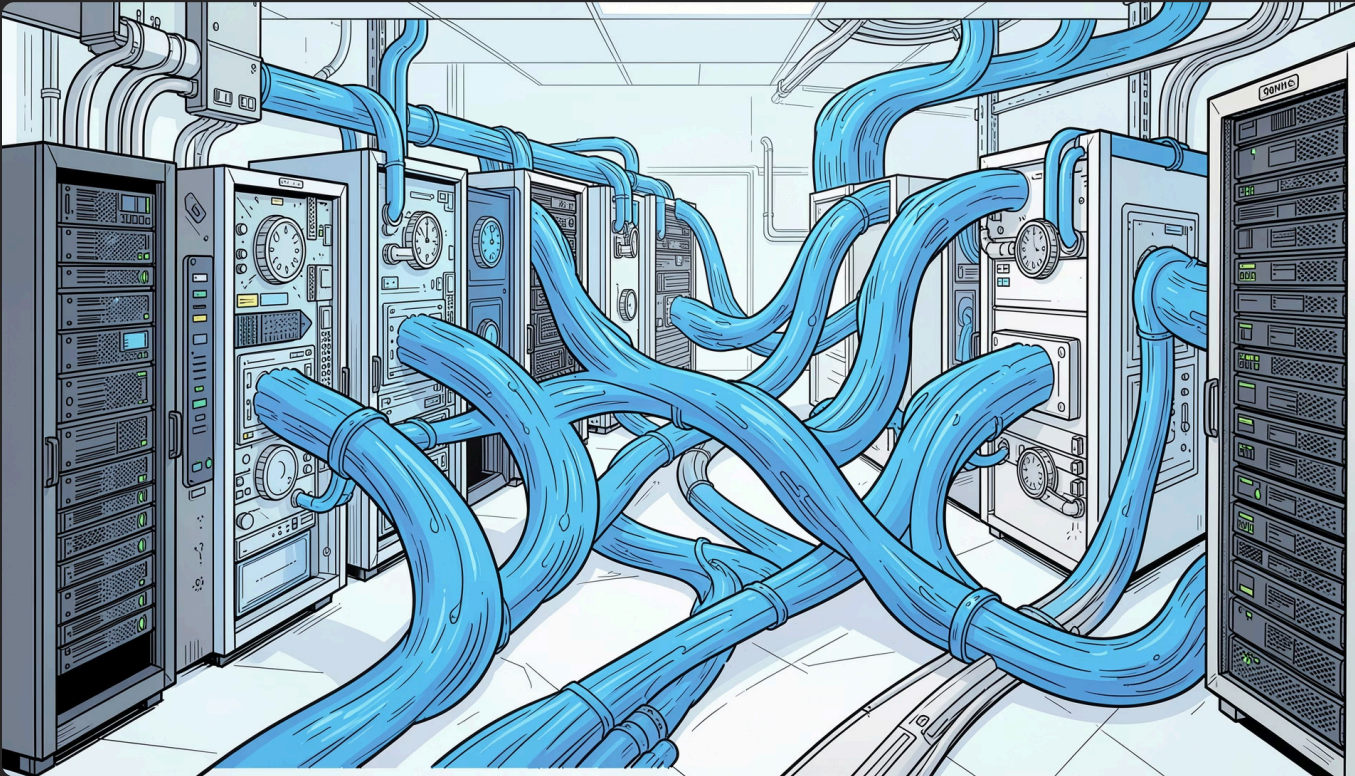
Credit Card & Customer Analytics

End-to-End Data Engineering & Business Intelligence Project Report

By : Ajay Kumar



Project Objective



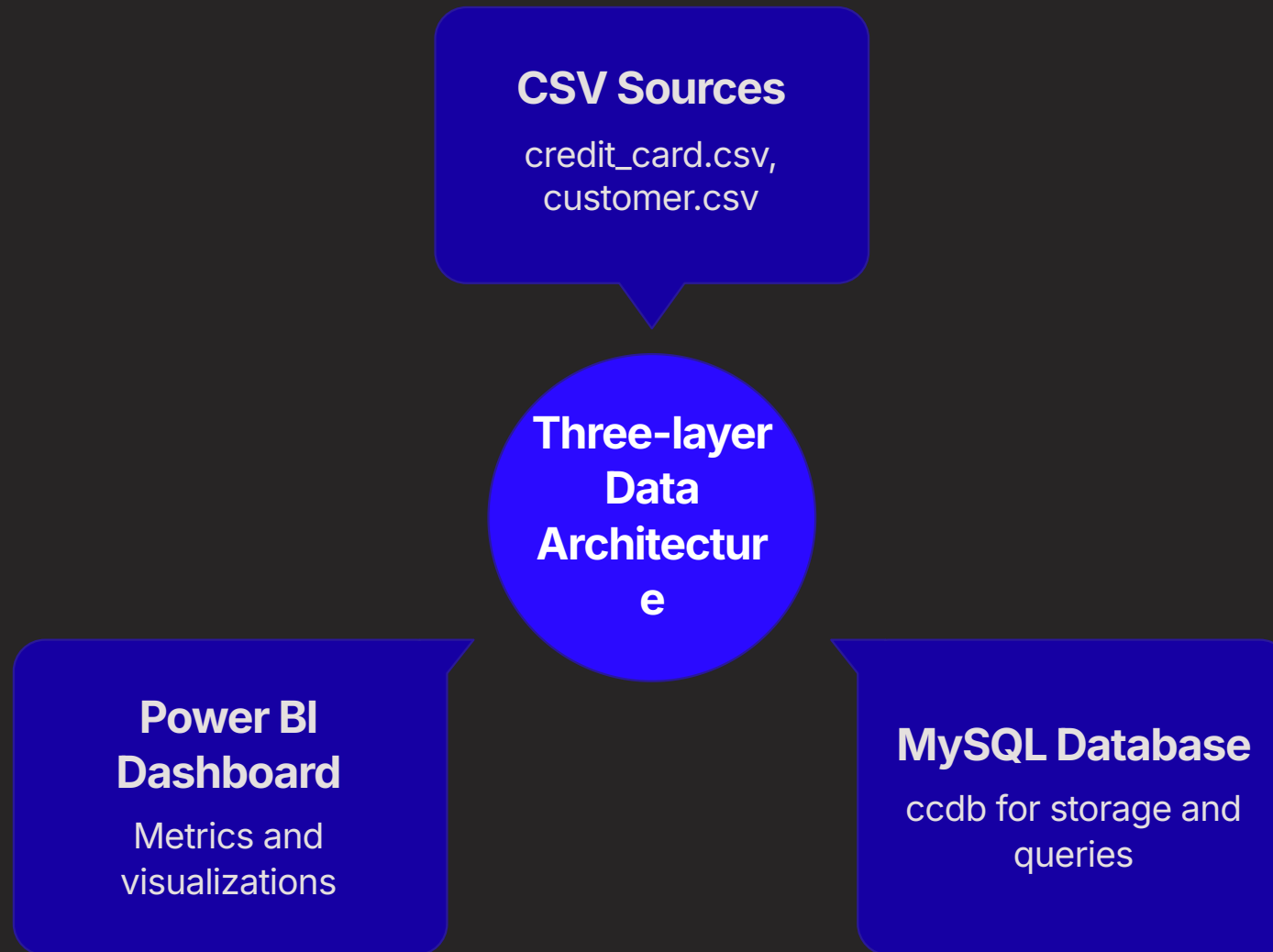
What We Built

A fully integrated pipeline transforming raw transactional records into actionable business intelligence.

What It Enables

- Monitor multi-million dollar asset portfolios
- Week-over-week growth tracking
- Early-warning asset health triggers

System Architecture



Optimized for performance, scalability, and seamless metric delivery across all pipeline stages.

Database Design

Schema: **ccdb**

Two core tables joined via `Client_Num` in a 1-to-Many relational mapping.

cc_detail

Transactions, fees, limits,
utilization, delinquency

cust_detail

Demographics, income,
job, satisfaction score

Ingestion Method

High-velocity `LOAD DATA LOCAL INFILE` commands with dynamic date parsing via `STR_TO_DATE()`.

- ① Designed to process millions of rows with full constraint execution and indexing.

YTD Portfolio KPIs

\$57M

Overall Revenue

\$46M

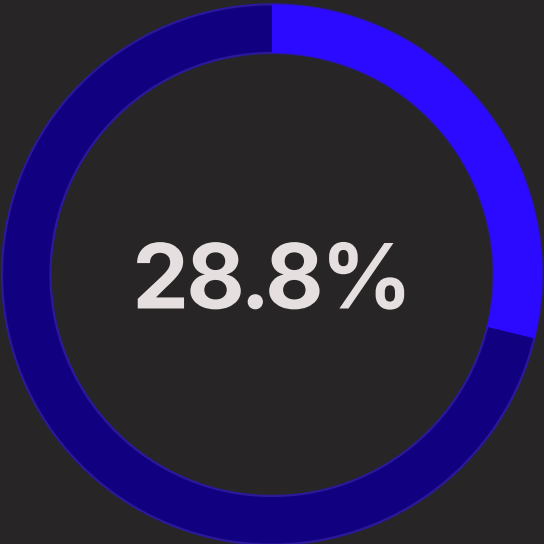
Transaction Amount

\$8M

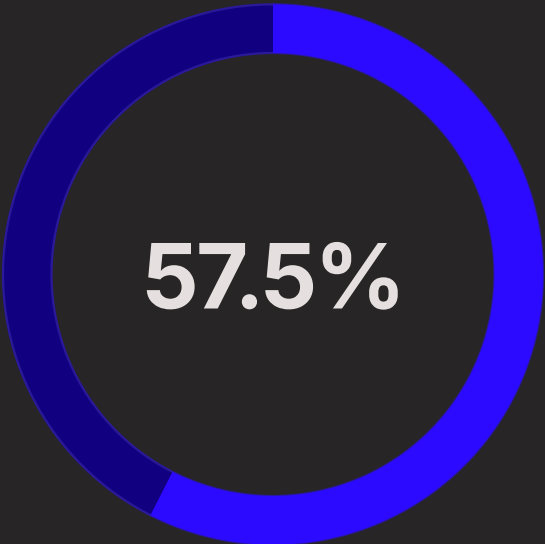
Total Interest

667K

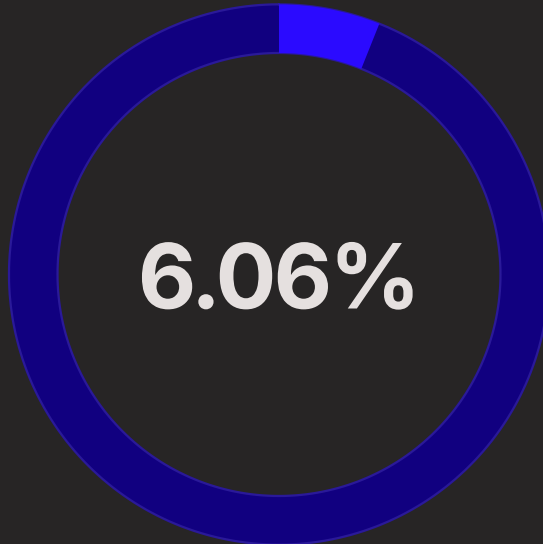
Transaction Count



WoW Growth

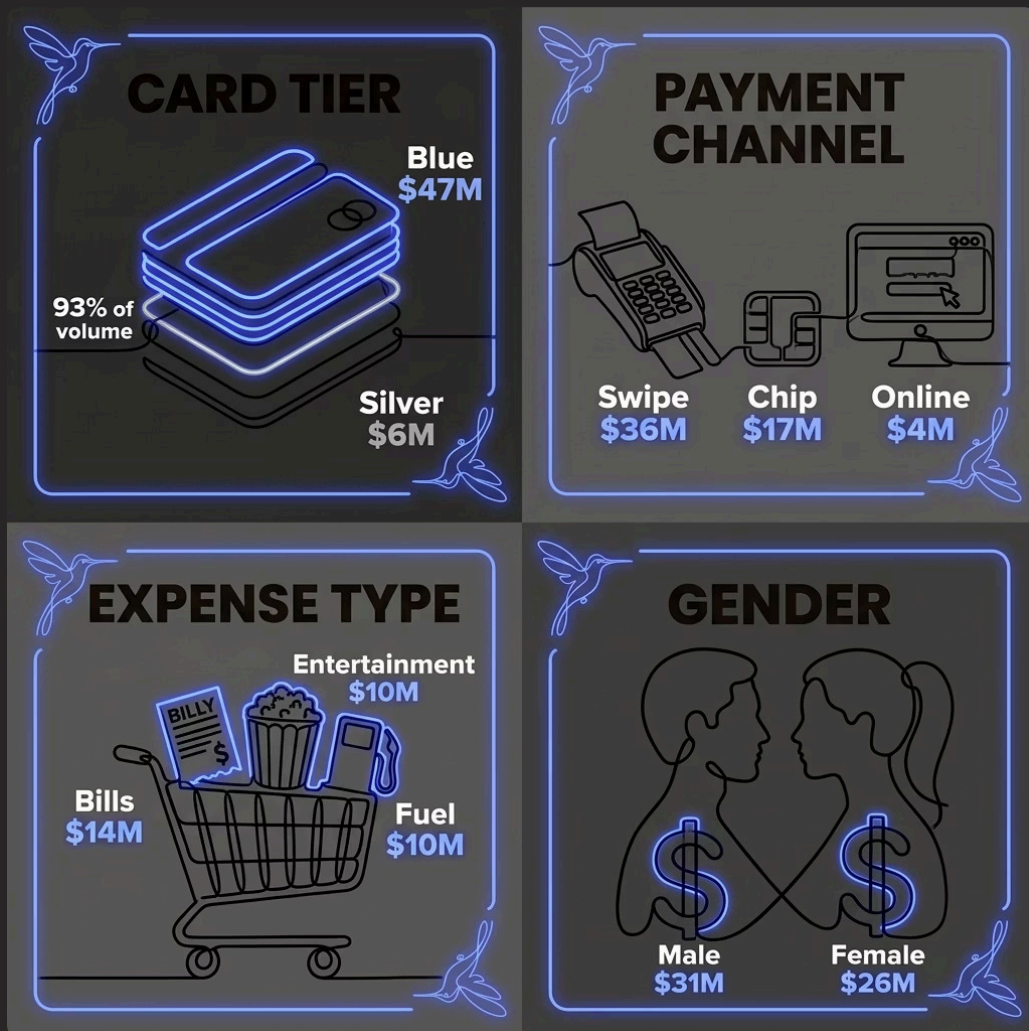


Activation Rate



Delinquency Rate

Multi-Dimensional Dashboard Analysis

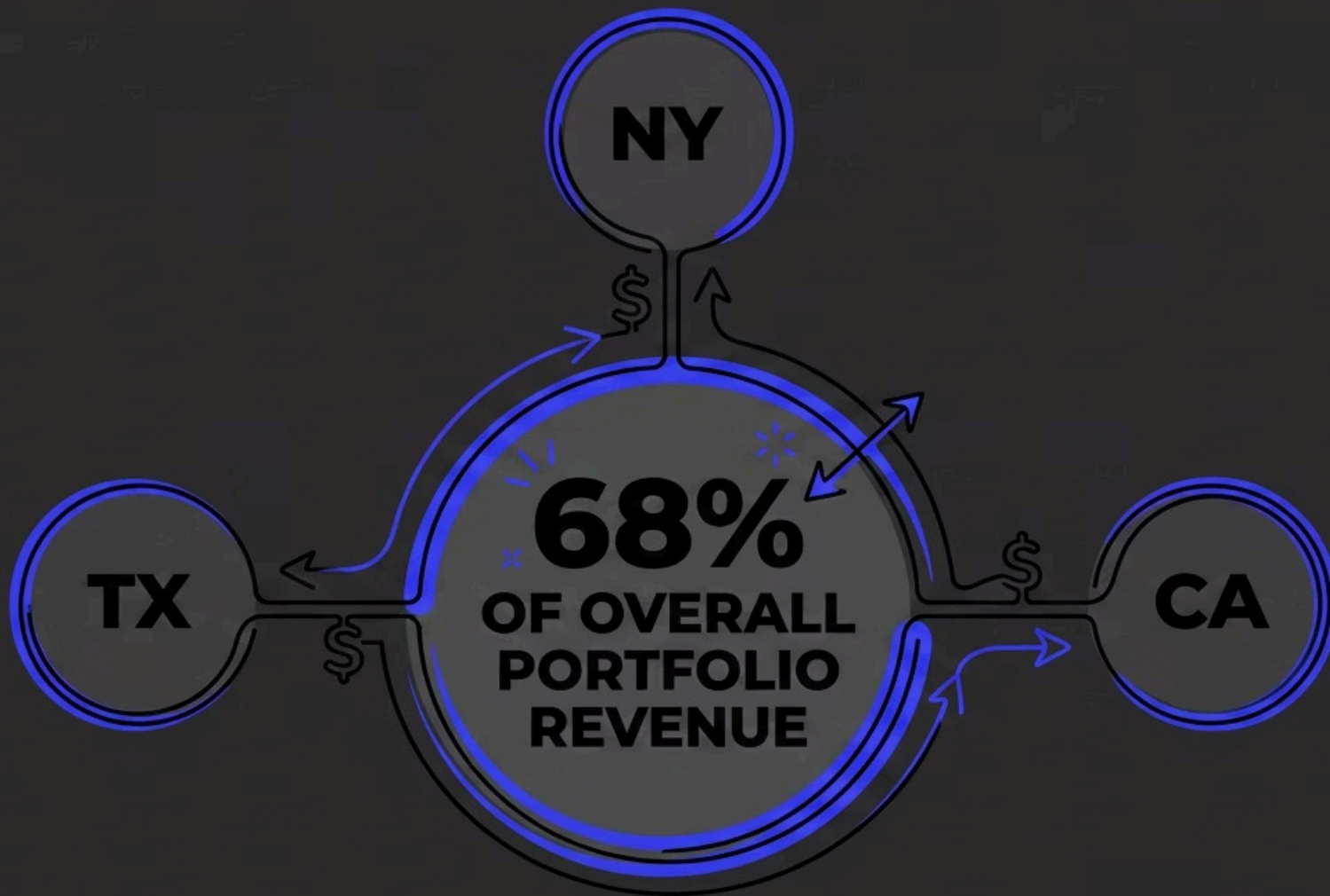


Key Findings

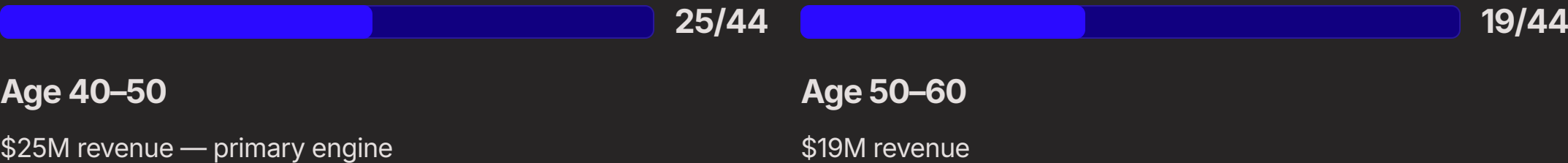
- **Blue + Silver** cards = 93% of all transactions
- **Swipe** dominates at \$36M; Online lags at \$4M
- **Bills, Entertainment, Fuel** are top spend categories
- **Male clients** generate \$31M vs. Female \$26M

Demographics & Geography

Top Revenue States



Age & Income Segments



Top Professions

- Business owners: \$18M
- White-collar professionals: \$10M

Strategic Recommendations

Boost Digital Channels

Online volume at \$4M vs. Swipe \$36M — launch virtual cards & e-commerce rewards.

Premium Tier Migration

93% in Blue/Silver — target high-income users for Gold & Platinum upsell campaigns.

Focus Regional Marketing

TX, NY, CA = 68% of revenue — concentrate spend, co-branded partnerships here.

Delinquency Early-Warning

6.06% delinquency rate — deploy predictive models flagging utilization spikes for automated intervention.

